

DEPTH_MD, DEPTH_TVD & INCLINATION - TRAJECTORY PARAMETERS

1. DATASET FIELDS REFERENCED IN THIS GUIDE

Field names below match the NWIS drilling_timeseries schema exactly, so this guide can be cross-referenced directly against logged data.

Field Name	Unit	Typical Range	Description
Depth_MD	m	0 - 3400 m	Measured depth: total length of wellbore drilled along its actual path
Depth_TVD	m	0 - 3350 m	True vertical depth: vertical distance from surface, ignoring wellbore path
Inclination	deg	0 - 42 deg	Angle of the wellbore from vertical at a given Depth_MD
Formation	text	Formation_W/X/Y/Z...	Named geological formation being penetrated at the current depth

2. HOW THESE FIELDS RELATE TO EACH OTHER

Depth_MD vs Depth_TVD: In a vertical well, Depth_MD approximately equals Depth_TVD. As Inclination increases in a directional well, Depth_MD grows faster than Depth_TVD for the same vertical progress, since the wellbore travels a longer slant path to reach the same true vertical depth.

Inclination vs Depth_MD: Inclination typically increases through a planned 'build section' (e.g. 0 to 28-42 deg over a defined Depth_MD interval), then holds roughly constant through the 'tangent' section - the build rate (deg/30m) is itself a derived risk factor for torque and hole cleaning.

Formation vs Depth_TVD: Formation tops/bases are typically correlated and prognosed against Depth_TVD (true stratigraphic position), not Depth_MD, since geological layering is a function of vertical position, not wellbore path length.

Inclination vs Hole Cleaning: Higher Inclination values increase the risk of a cuttings bed forming on the low side of the hole, which raises Torque and reduces effective annular velocity for a given Flow_Rate.

3. KEY FORMULAS / RULES OF THUMB

Formula / Rule	Notes
$\text{Depth_MD} - \text{Depth_TVD} = \text{cumulative slant offset from vertical}$	A growing gap between the two fields over a well's depth log is a quick check that the well is deviating as planned.
$\text{Build Rate (deg/30m)} = \text{Change in Inclination} / (\text{Change in Depth_MD} / 30)$	Rates above ~4 deg/30m are associated with elevated dogleg-related torque risk (see Field_Alpha cross-reference).
Dogleg Severity combines Inclination AND azimuth change per 30m interval	Depth_MD, Depth_TVD, and Inclination logged together let this be reconstructed even without a stored azimuth field.

4. RISK INDICATORS & RECOMMENDED ACTIONS

Observed Pattern (Fields Involved)	Risk Indicated	Recommended Action
Inclination increasing rapidly (build rate >4 deg/30m) at a given Depth_MD	Elevated dogleg severity - torque/drag and key-seat risk	Flag the interval for reduced ROP/WOB and increased hi-vis sweep frequency; alert directional driller to confirm planned build rate

Depth_MD progressing with Depth_TVD nearly flat	Wellbore is in a high-inclination or near-horizontal section	Increase Flow_Rate or sweep frequency, since standard vertical-well hole-cleaning assumptions no longer apply
Formation change logged without an update to Mud_Weight/Formation_Pore_Pressure assumptions	Pressure or lithology-related risk may be unaccounted for	Cross-check the new Formation against its pore-pressure/fracture-gradient prognosis before continuing

5. SAMPLE LOG ROWS (drilling_timeseries SCHEMA)

Well_ID	Timestamp	Depth_MD	Depth_TVD	Inclination	Formation
W002	2024-10-02 04:00	3148	2967	28.0	Formation_Y (lower)
W005	2025-01-24 04:00	2740	2412	31.5	Formation_W2 (lower)

6. FIELD_ALPHA CROSS-REFERENCE

- W002 (Alpha-02): build 0-28 deg completed by Depth_MD 2600 m; the 28-deg tangent section (Depth_MD 2600-3320 m, Formation_Y) is where the stuck-pipe event at 3160 m occurred.
- W005 (Alpha-05): a concentrated dogleg cluster (4.2 deg/30m) around Depth_MD 2730-2750 m in Formation_W2 directly contributed to the torque spike at 2740 m.

7. OPERATIONAL BEST PRACTICES

- Always log Depth_MD, Depth_TVD, and Inclination together - Depth_MD alone cannot distinguish a vertical well from a highly deviated one.
- Compute build rate as a derived field from consecutive Inclination/Depth_MD readings to catch dogleg clusters automatically.
- Tie Formation boundaries to Depth_TVD in the reference model, but monitor drilling risk (torque, hole cleaning) against Depth_MD/Inclination.

This Drilling Parameter Reference Guide is a synthetically generated reference document produced for the NWIS prototype, aligned to the drilling_timeseries dataset schema. Not associated with an actual drilling operation.